**Pure Year 2 · Chapter 8**

# Parametric Equations

Cartesian forms · Sketches · Intersections · Modelling

## Starter 2

1. Make $t$ the subject:

a. $x = 4t - kt$

b. $y = 3t^2$

c. $y = 2 - 4\ln t$

d. $x = 1 + 2e^{-3t}$

1

2. Express using cosine:

a. $4 + 3\sin^2 x$

b. $\sin 2x$

c. $\cot x$

d. $2\cos x + \cos 2x$

2

3. Find the ranges:

a. $\ln(x+1),\ x > 0$

b. $2\sin x,\ 0 < x < \pi$

c. $x^2 + 4x - 2,\ -4 < x < 1$

d. $1/(2x+5),\ x > -2$

3

4. Find the intersections of the circle with centre $(0, 4)$, radius $5$, and the line $2y - x - 10 = 0$.

4

## Chapter Overview 3

**Chapter overview**

1. Convert $x = 2\cos t + 1,\ y = 3\sin t$ to Cartesian form.
2. Sketch $x = 2t,\ y = 5/t$.
3. Find where $x = t^2,\ y = 4t$ meets $x + y + 4 = 0$.
4. Model a plane using $x = (v\cos\theta)t,\ y = (v\sin\theta)t$.

## What are they and what is the point? 4

A Cartesian equation relates $x$ and $y$, for example $x^2 - y^2 = 5$.

Parametric equations express both coordinates using another variable, such as $x = \sin t,\ y = t^2$.

Each allowed value of $t$ gives a point on the curve. Eliminating $t$ does not always preserve the full domain or direction.

## Converting parametric to Cartesian 5

Find a Cartesian equation and the domain and range of

$x = 2t, \quad y = t^2, \quad -3 < t < 3.$

Answer

Sketch

## Further Example 6

Find a Cartesian equation and state its domain and range:

$x = \ln(t + 3), \quad y = 1/(t + 5), \quad t > -2.$

Answer

## Now you try. 7

1. Eliminate $t$: $x = \ln(t+2),\ y = 1/(t+1),\ t > -1$.

1

2. Eliminate $t$: $x = \ln t,\ y = t^2 - 2,\ t > 0$.

2

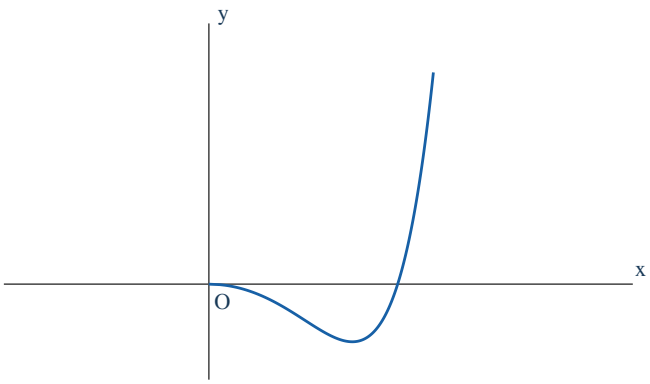
## Pure Year 2/AS Exercise 8A (200-202) 8

8. A curve is given by
   $x = 3\sqrt{t},\ y = t^3 - 2t,\ 0 \le t \le 2$.

a. Find its Cartesian equation and domain.

b. Find where $dy/dt = 0$.

c. Hence find the range of $y$.




a

b

c

9. $x = t^3 - t,\ y = 4 - t^2,\ t \in \mathbb{R}$.

a. Show that $x^2 = (a - y)(b - y)^2$, finding $a, b$.

b. Find the greatest value of $y$.

a

b

## . . .when you have trig identities 10

1. Eliminate $t$ from $x = \sin t + 2,\ y = \cos t - 3$.

1

2. Eliminate $t$ from $x = \sin t,\ y = \sin 2t,\ -\pi/2 \leq t \leq \pi/2$.

2

Sketch

## Now you try.. 11

1. $x = 2\sin t,\ y = 1 - \cos 2t,\ -\pi/2 \le t \le \pi/2$. Find a Cartesian equation and its domain.

1

2. $x = \cot t + 2,\ y = \csc^2 t - 2,\ 0 < t < \pi$. Find a Cartesian equation and sketch the curve.

2

Sketch

## Exercise 8B Pure 2 pages 204-206 12

6. $x = 8\cos t,\ y = \frac{1}{4}\sec^2 t,\ -\pi/2 < t < \pi/2.$

Find a Cartesian equation, its domain and range. The endpoints $t = \pm\pi/2$ are excluded because $\sec t$ is undefined.

Answer

7. $x = 3\cot^2 2t,\ y = 3\sin^2 2t,\ 0 < t \leq \pi/4$. Find a Cartesian equation and its domain.

Answer



8. $x = \frac{1}{3}\sin t,\ y = \sin 3t,\ 0 < t < \pi/2$.

a. Show $y = ax(1 - bx^2)$, finding $a, b$.

b. State the domain and range.

a

b

10. $x = \tan^2 t + 5,\ y = 5\sin t,\ 0 < t < \pi/2.$

a. Find a Cartesian equation.

b. State the domain and range.

a

b



11. $x = \tan t,\ y = 3\sin(t - \pi),\ 0 < t < \pi/2.$ Find a Cartesian equation and its domain and range.

Answer

**Sketching a spiral**

$x = \theta\cos\theta, \quad y = \theta\sin\theta, \quad 0 \le \theta \le 4\pi.$

Mark the points for $\theta = 0, \pi/2, \pi, 3\pi/2, 2\pi$. What happens during the second turn?

Sketch

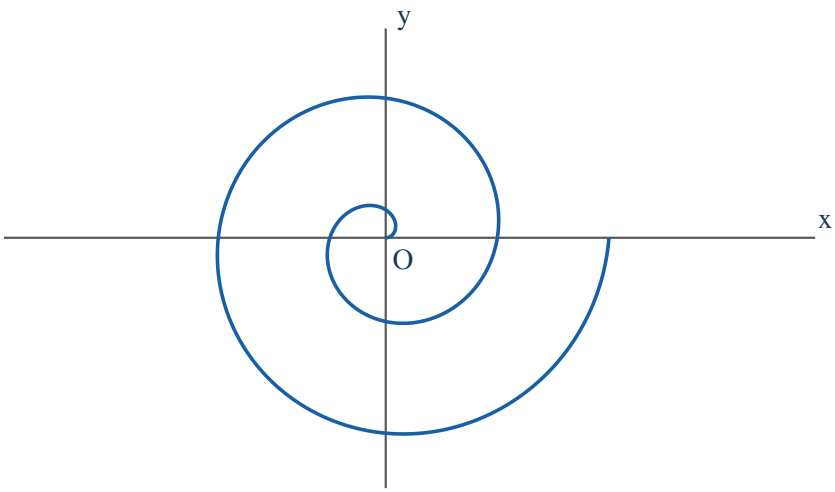
Sketch



**Plot points in parameter order**

| $\theta$ | $0$ | $\pi/2$ | $\pi$ | $3\pi/2$ | $2\pi$ |
|---|---|---|---|---|---|
| $x$ | $0$ | $0$ | $-\pi$ | $0$ | $2\pi$ |
| $y$ | $0$ | $\pi/2$ | $0$ | $-3\pi/2$ | $0$ |

For $x = \theta\cos\theta,\ y = \theta\sin\theta$, plot further points up to $4\pi$. Join them smoothly and show the direction as $\theta$ increases.



## Now you try.. 19

Sketch $x = 2t,\ y = t^2,\ -1 \leq t \leq 5$, showing endpoints and direction.

Sketch

Sketch

## Exercise IM Level 1 Exercise 1 20

Find Cartesian equations:

i. $x = 1 - t,\ y = t^2 - 4$

ii. $x = 2t^2,\ y = 1/t, t \neq 0$

iii. $x = 2\cos\theta + \sin\theta,\ y = \cos\theta - 2\sin\theta$

i

ii

iii

## Exercise IM Level 1 Exercise 2 21

For $x = t^2,\ y = t^3$, complete a table for integer $t$ from $-3$ to 3, sketch the curve and find a Cartesian equation.

Table

Equation

Sketch

## Exercise 8C (Pages 207-208) 22

Find the Cartesian equation and sketch $x = 3 - t,\ y = t^2 - 2,\ -2 \leq t \leq 3$.

Answer

Sketch

## Exercise 8C (Pages 207-208) 23

$x = 9\cos t - 2,\ y = 9\sin t + 1,\ -\pi/6 \le t \le \pi/2.$

a. Find a Cartesian equation.

b. Sketch the arc, giving its endpoints.

c. Find its length.

a

b

c

Sketch

## Points of Intersection 24

$x = at^2 + t,\ y = a(t^3 + 8)$ passes through $(-4, 0)$. Find $a$ and the intersections with the $y$-axis.

Find a

Intersections

## Points of Intersection 25

1. Find where $x = t^2,\ y = 4t$ meets $x + y + 4 = 0$.

1

2. $x = \cos t + \sin t,\ y = (t - \pi/6)^2,\ -\pi/2 < t < 4\pi/3$. Find points with $y = \pi^2$, and intersections with the $y$-axis.

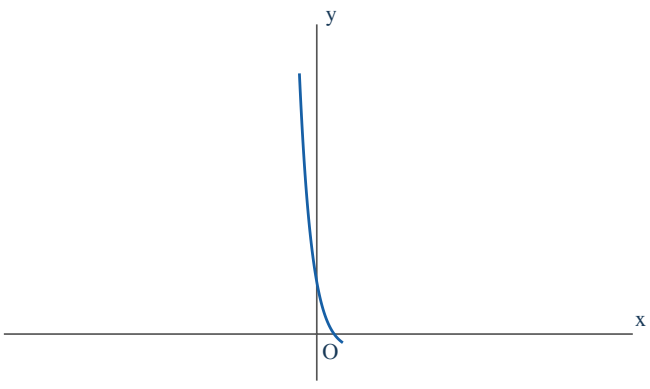
2: given y

2: axis

## Now you try 26

*C4 January 2013*

$x = 1 - t/2,\ y = 2^t - 1$. Find the intercepts with both axes.




y-axis

x-axis

3. $x = 4at^2,\ y = a(2t - 1)$ passes through $(4, 0)$. Find $a$.

Answer



4. $x = b(2t - 3),\ y = b(1 - t^2)$ passes through $(0, -5)$. Find $b$.

Answer

5. Find the intersection of $x = 3t + 2,\ y = 1 - t$ with the line $x + y = 2$.

Answer

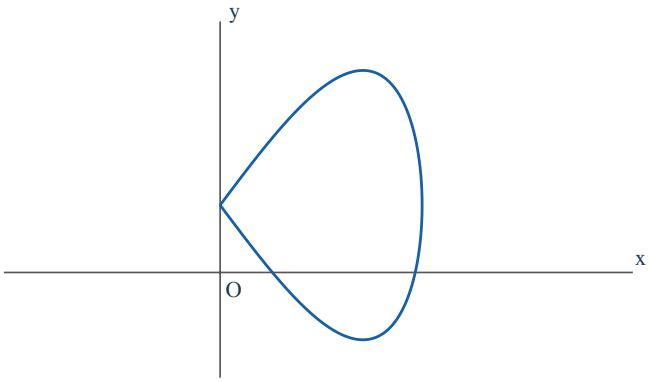


12. $x = 6\cos t, y = 4\sin 2t + 2,$
$-\pi/2 < t < \pi/2.$

a. Find the $x$-axis intercepts.

b. Find the parameter values for intersections with $y = 4$.

c. Give the coordinates of these points.




a

b

c

## Exercise 8D (page 211-213) Q13 31

Show that the line $y = 2x - 5$ does not intersect $x = 2t,\ y = 4t(t-1)$.

Answer

## Exercise 8D (page 211-213) Q14 32

$x = \sin t,\ y = \cos 2t + 1,\ 0 \leq t \leq 2\pi$.

a. For which values of $k$ does $y = k$ meet the curve?

b. Find a Cartesian equation and the horizontal tangent.

a

b

Sketch

## Modelling 33

A plane travels at constant speed $v$, making angle $\theta$ with the horizontal. It rises 120 m for every 600 m horizontally.

a. Find $\theta$.

b. Give parametric equations when $v = 50\,\text{m}\,\text{s}^{-1}$.

c. Find its height after 10 seconds and a Cartesian equation. Comment on the model.

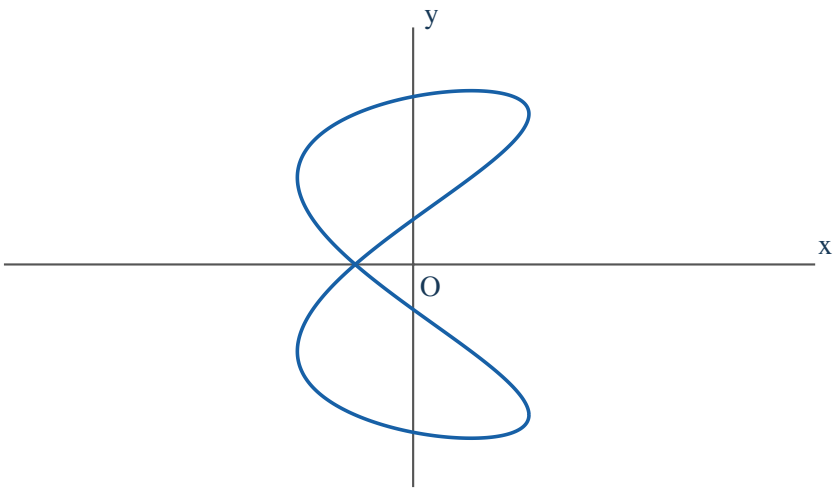
a

b

c

## Further Example 34

A skater follows $x = 8\cos 20t,\ y = 12\sin(10t - \pi/3)$, where $t$ is in minutes. Find the initial point, the self-intersection, the $y$-axis crossings and the time for a full circuit.




Initial point

Crossings

Circuit

1. A boat crosses a 75 m wide river, with eastward distance $x = 0.9t$ m and northward distance $y = -3.2t$ m, after $t$ seconds.

a. Find the crossing time.

b. Find the southward drift.

c. Show the path is straight.

d. Find its speed.

a

b

c

d



2. A descending plane is modelled by $x = 80t,\ y = 3000 - 9.1t$, with distances in metres and $t$ in minutes.

a. Find its initial height.

b. Discuss the stated domain $0 \le t \le 329$.

c. Find the horizontal distance at landing.

a

b

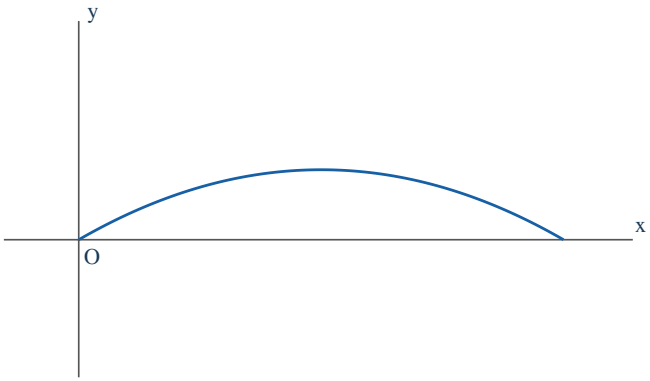
c

3. A ball is kicked from the ground at $20\,\text{m s}^{-1}$, angle $30^\circ$. Its path is $x = 10\sqrt{3}t,\ y = 10t - 4.9t^2$.

a. Find its horizontal range.

b. Find when it is descending between heights 1.5 m and 2.5 m.

c. Find the nearest point where a player can head it in that interval.




a

b

c



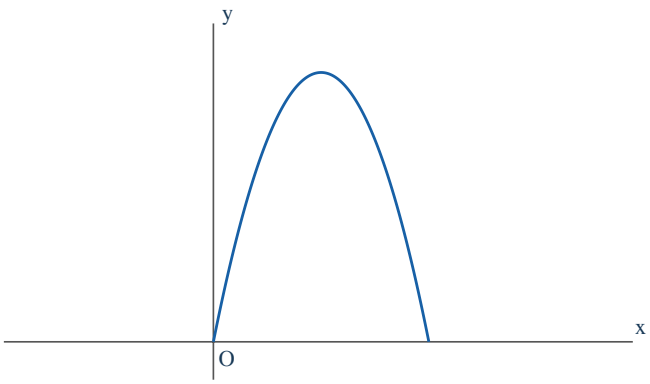
4. A dolphin leaps along $x = 2t,\ y = 10t - 4.9t^2$, in metres and seconds.

a. Find the jump duration.

b. Find its horizontal distance.

c. Show the path is quadratic.

d. Find its maximum height.




a

b

c

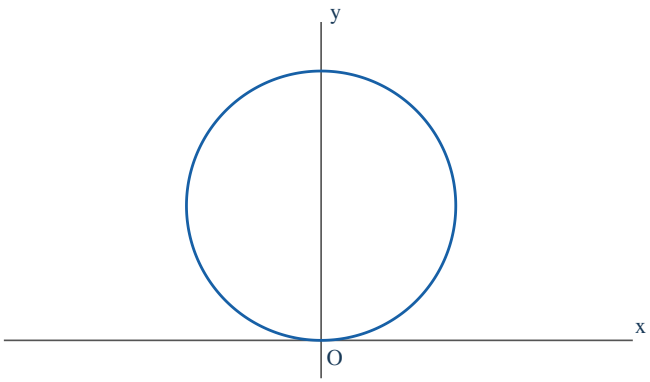
d

5. A Ferris-wheel car has $x = 12\sin t,\ y = 12 - 12\cos t$, distances in metres and $t$ in minutes.

a. Show its path is a circle of radius 12 m.

b. Find its maximum height.

c. Find the period and average speed.




a

b

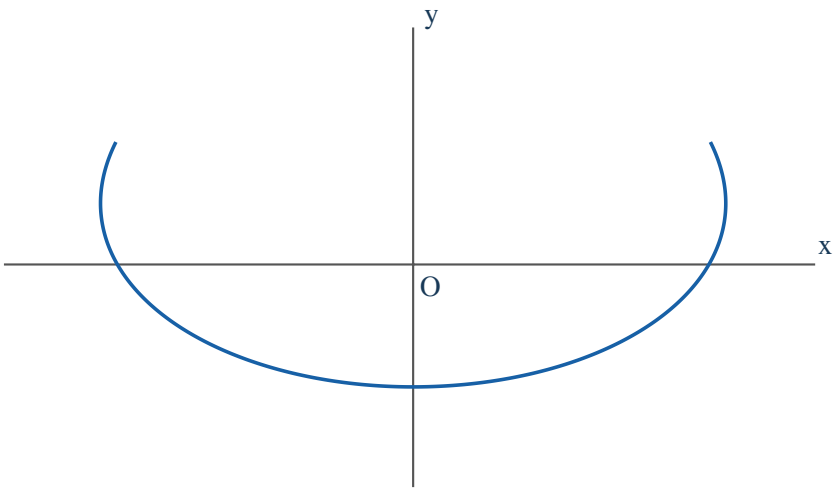
c



6. A bowl has cross-section $x = t - 4\sin t,\ y = 1 - 2\cos t,\ -\pi/2 \le t \le \pi/2$.

a. Find the width of the opening.

b. Given that the deepest point is on the $y$-axis, find the depth.




a

b

**Explore an implicit curve**

$\sin(\sin x + \cos y) = \cos(\sin(xy) + \cos x)$.

Not every curve has a convenient explicit or parametric representation. A contour plot draws the zero set of the difference between the two sides.

